

COMPSYS704 – Advanced Embedded Systems Project 1 Descriptions and Hints

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Main goals and requirements

- Advantech is a company for manufacturing very sensitive and expensive liquids for their customers per custom order
- The process has two stages:
 - Producing liquids and temporary storage in special tanks
 - Bottling liquids selected in purchase order and storage in strict climate and security conditions before delivery
- Advantech introduces a number of systems in their new facility which use IIoT technologies
- Their goal is to be Industry 4.0 compliant

Advantech's new facility

- 2 sections: office and manufacturing
- 7 physical zones equipped with different kinds of sensor nodes, sensors and actuators which can be accessed over networks
- In this project we focus on Zone 5 within manufacturing section

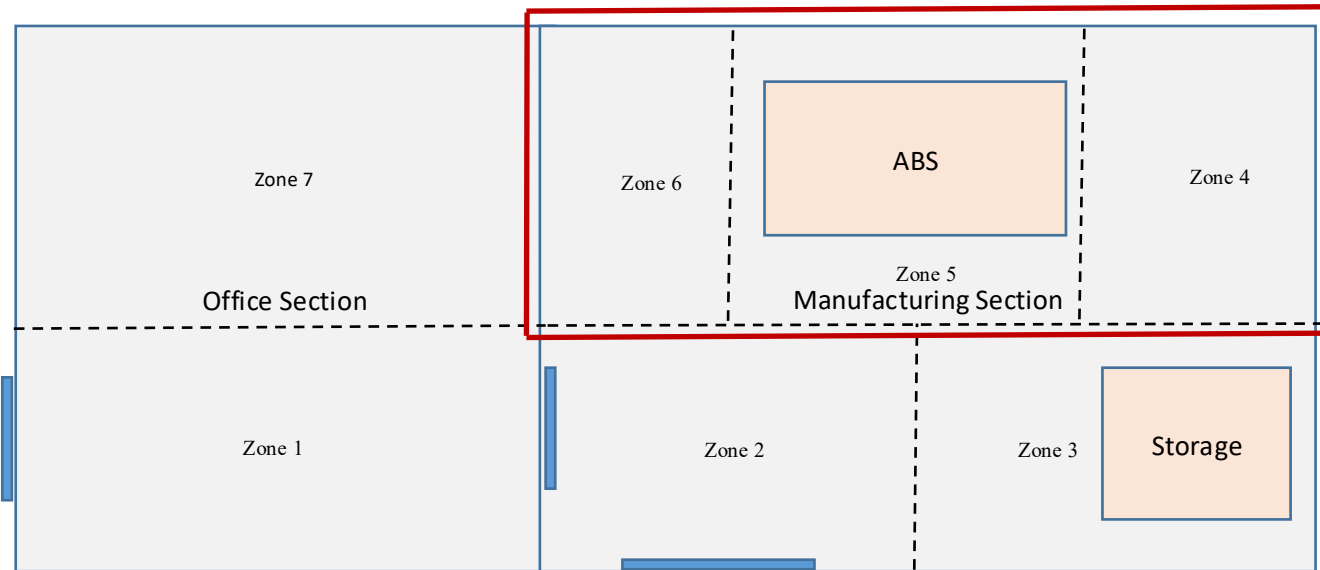
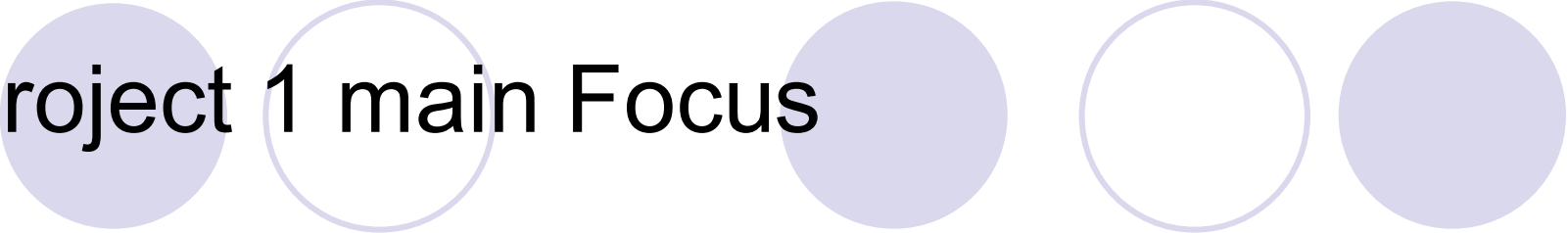


Figure 1 Manufacturing facility conceptual layout

Project 1 main Focus



- Group Project:

- Automated Bottling System, ABS – Group Project
- Simple Purchase Order System, POS – Group Work
- Must be partitioned between members of the group

- Individual project:

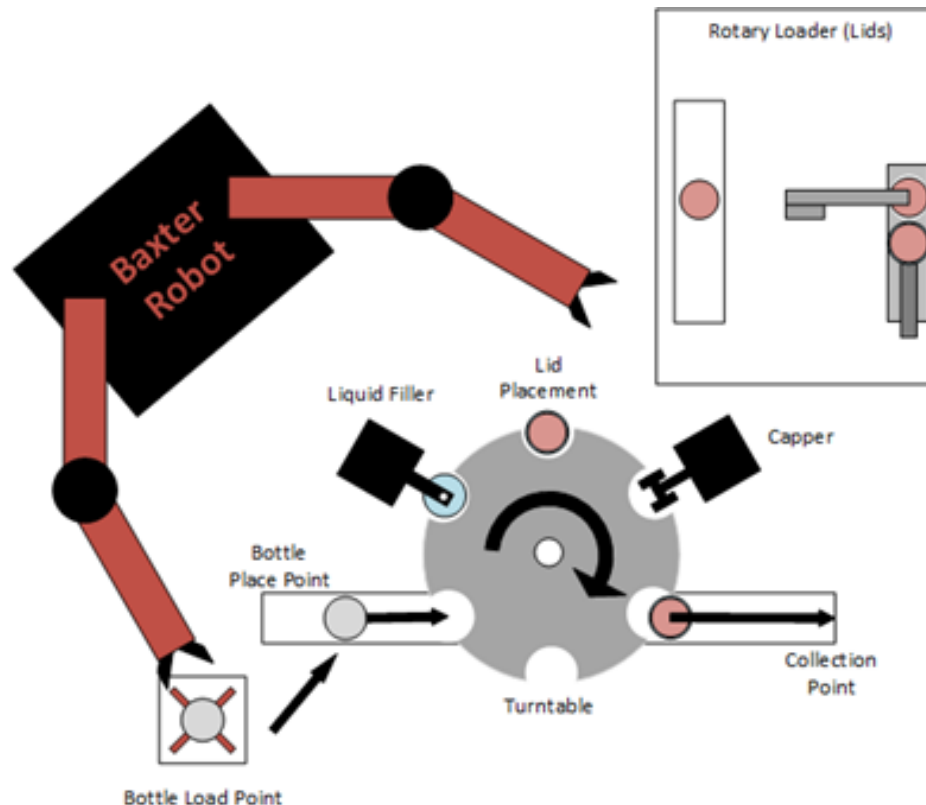
- Advanced functions/features added to ABS (EABS)

System requirements (look to IRP brief)

- ABS has to be able to process multiple bottles at the same time (production pipeline)
- Introduce assumptions (two different liquids and how many different combinations in PO; quantities of each)
- POS has user entry/PO completion and launch to the factory; you may try with different computers that run ABS and POS
- Other advanced features – explored and included through IP (Digital Twins, fault-tolerance, reconfigurability of the production process with respect to products and machine/workstation availability and status)

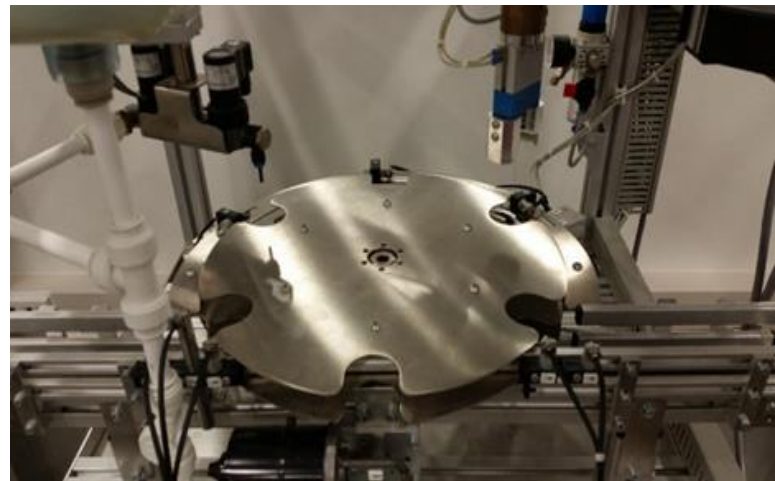
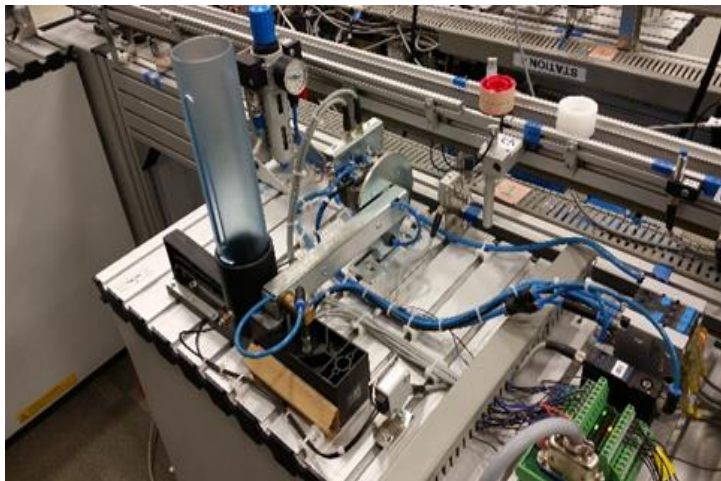
Automated Bottling System

- Robot or pick and place arm added to the ABS
- Abstracted loading of bottles on Conveyor; Lid loader is used to load caps on bottles



Automated Bottling System

- Multiple intelligent machines have to be designed and integrated into the ABS
 - Models of controllers and plants need to be developed for simulation (excludes the Cap/Lid loader which is provided through the Lab 3)
 - Capper can be simplified

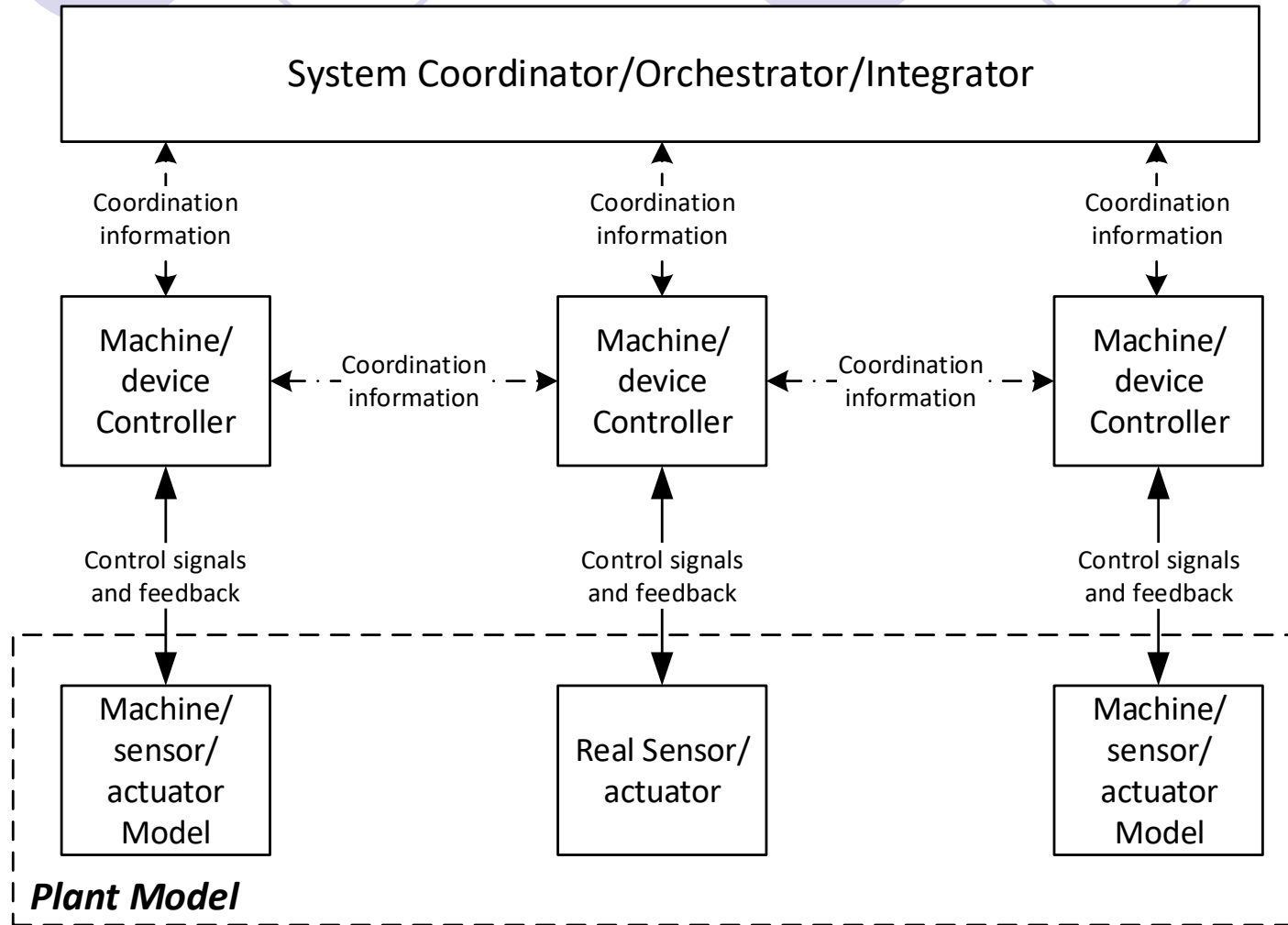




Purchase Order System

- Presents a purchase order form to the customer
- Accepts specifications for up to four different products (with % of each liquid, totaling 100%)
- Accepts volume (number of instances of products of the same type)
- Sends the message with the new order to (E)ABS
- Receives information of completion time from (E)ABS

Possible ABS Architecture



Individual Project themes

Some ideas listed in following slides

- You need to make your own (meaningful) specifications and design
- Make assumptions as you see fit – talk with teaching team if in doubt
 - Make simple visualization of the overall system (accurate state has to be captured)
 - Digital Twins of workpieces (WP), workstations (WS) and Transport Devices (TD)
 - Fault-tolerance,
 - Tracking of WPs
 - Reconfiguration of production procedure (recipe)
 - More....all proposals will be considered and if sound approved

Tentative Milestones (follow announcements and lectures)

Tentative Milestone	Description	Expectations
M1, week 6 GRP-> 10/40 marks IRP-> 5/20 marks	Preliminary specification and conceptual design of the whole system completed; one machine (plant + controller) model completed, preferably turntable or single-liquid filler Conceptual design of IRP selected system made and interface and integration into system described	The whole design must be specified as conceptual design and submitted for review. It can be in the form of diagrammatic representation of the system with main interfaces between the parts and rough explanation of functionality of parts and simple descriptions. The design of one intelligent machine model from ABS (any except workpiece loader) has to be completed, including visualisation. In IRP main selection of the system feature/topic selected and described. More in Project 1 brief.
M2, week 7 (after the study break) GRP->30/40 marks IRP->15/20 marks	Final design	EABS completed. Visualisation completed. Presentation of the project to the class/teaching team. Deliverables described in Project 1 brief. through Canvas (TBA, include the ABS, overall design diagram, short descriptions of the solution). Working selected topic of IRP described, designed and prepared for presentation.